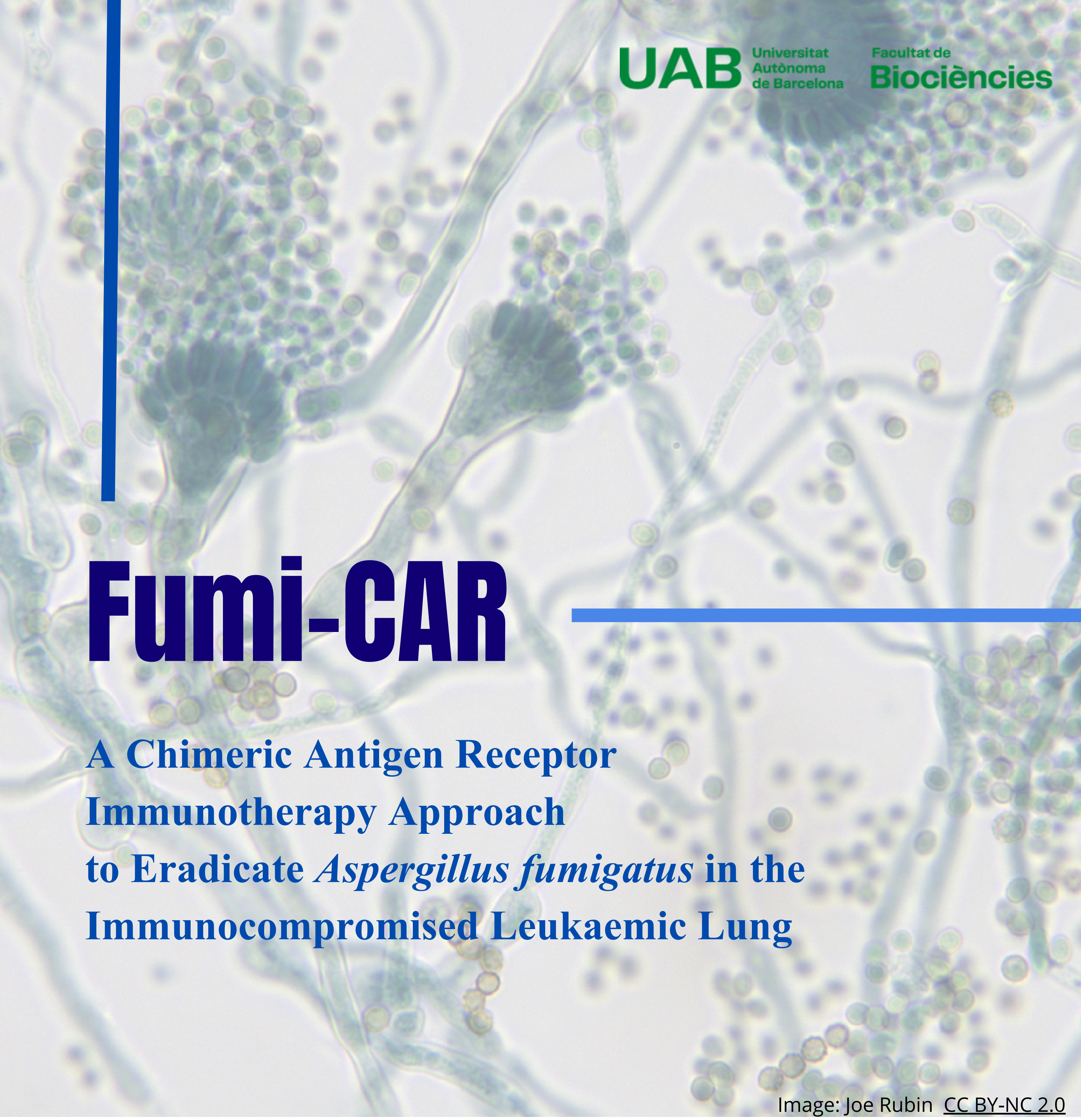

UAB Universitat Autònoma de Barcelona
Facultat de Biociències

Fumi-CAR

A Chimeric Antigen Receptor Immunotherapy Approach to Eradicate *Aspergillus fumigatus* in the Immunocompromised Leukaemic Lung

Bachelor's thesis - Research project proposal

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Abstract

Invasive Pulmonary Aspergillosis (IPA) remains a devastating complication of acute leukaemia (AL), affecting 5–6% of patients with mortality rates exceeding 40% despite antifungal therapy. Current treatments are limited by toxicity, drug interactions, and rising azole resistance, while the profound immune failure characteristic of AL renders conventional adoptive T-cell strategies largely ineffective. This project proposes Fumi-CAR, a fourth-generation TRUCK platform engineered to target *Aspergillus fumigatus* hyphae using an scFv derived from the preclinically validated AB90-E8 antibody. Manufactured from patient-derived T cells via lentiviral transduction, the construct incorporates a CD8α hinge, 4-1BB costimulation to preserve metabolic fitness and resist exhaustion, CD3ζ primary signaling, and an NFAT-inducible IL-12 cassette designed to recruit and activate residual innate immunity at the site of infection. This MHC-independent approach directly addresses the multi-layered immune paralysis of neutropenic AL patients. Antifungal efficacy will be evaluated through antigen-specific activation assays, direct hyphal damage quantification, and macrophage recruitment experiments *in vitro*, followed by a pilot *in vivo* study in an NSG cyclophosphamide-induced neutropenia model of IPA. If successful, Fumi-CAR could reduce fungal burden, limit tissue invasion, and improve survival where current therapies fail, establishing CAR technology as a viable immunotherapeutic platform for infectious disease beyond oncology.

Keywords: CAR T-cell therapy · *Aspergillus fumigatus* · Invasive Pulmonary Aspergillosis · Acute leukaemia · Antifungal immunotherapy

Contents

1. Introduction

Aspergillosis is a mycotic illness often induced by ***Aspergillus fumigatus**,* a saprophytic mould responsible for a significant global burden of **invasive fungal disease**, with an estimated 300,000 life-threatening infections annually. The dissemination of its small (2–3 µm) **conidia** is a universal environmental phenomenon, with daily inhalation constituting a constant, low-level exposure.[1]

As shown in Figure 1, in the immunocompetent individual, a coordinated, multi-faceted immune response integrating mucociliary clearance, alveolar macrophage phagocytosis, neutrophil extracellular traps (NETs), and a critical T-helper type 1 (Th1) adaptive response, ensures efficient eradication of these conidia.[2] However, in the **immunocompromised host**, inhaled *A. fumigatus* conidia that are not cleared at the airway surface can germinate within the pulmonary parenchyma, leading to the development of germ tubes and hyphae that invade lung tissue and, in severely neutropenic or T-cell–depleted hosts, penetrate blood vessels, causing angio-invasive disease, known as **Invasive Pulmonary Aspergillosis** (IPA).[3] From there, extrapulmonary dissemination to sites such as the sinuses and brain can occur, resulting in **Invasive Aspergillosis** (IA), the most severe manifestation of this disease.[4]

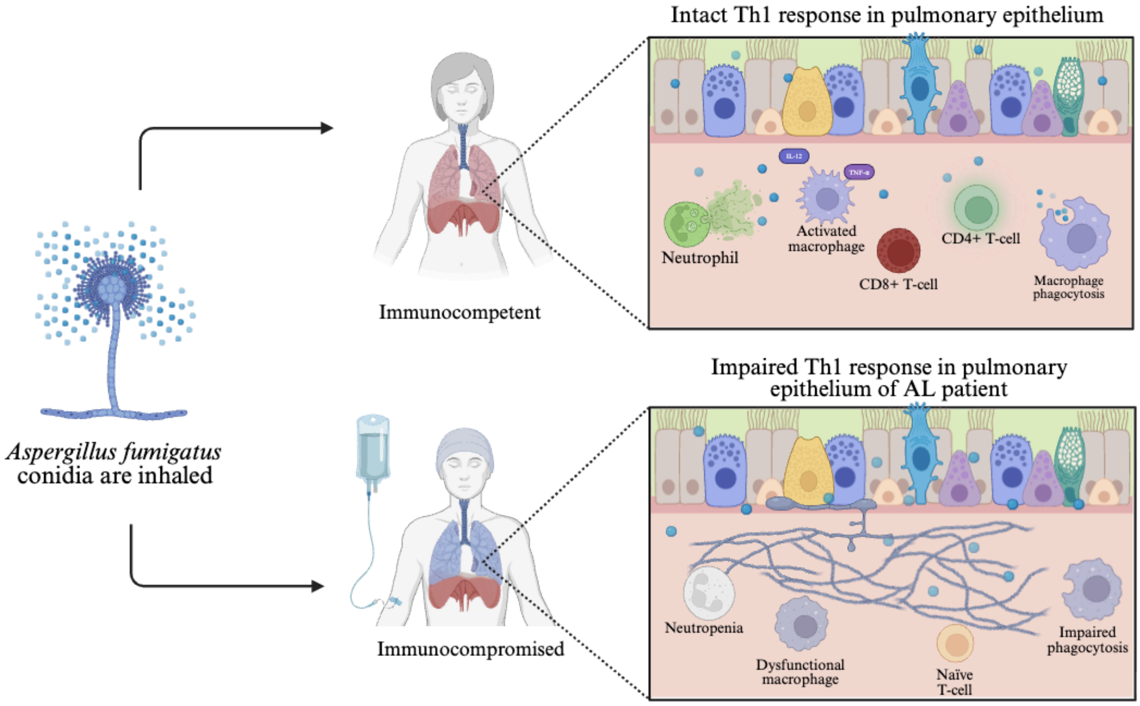

Figure 1. **Immune dysfunction in acute leukaemia patients predisposes to Invasive Pulmonary Aspergillosis**. In an immunocompetent host (upper), inhaled *A. fumigatus* conidia are effectively cleared by coordinated actions of alveolar macrophages, neutrophils, and Th1-polarized $CD4^+$ and $CD8^+$ T-cells. In contrast, patients with acute leukemia (lower) exhibit neutropenia, dysfunctional macrophages, and naive T-cells with impaired Th1 responses, leading to failed phagocytosis, uncontrolled hyphal growth, and angioinvasion. *Created using Biorender.*

Susceptibility is profoundly influenced by the severity of the immune status of the patient; patients with hematological malignancies, particularly **acute leukaemia** (AL), as well as recipients of hematopoietic stem cell transplantation, those exposed to intensive chemoradiotherapy, high-dose corticosteroids, or other potent immunosuppressants, constitute the archetypal high-risk population.[5]

In AL patients specifically, severe and sustained neutropenia impairs the principal defence against hyphae: **neutrophils**.[6] Compounding this, **leukaemic blastaemia** directly inhibits the antifungal activity of surviving peripheral blood mononuclear cells, causing functional immune paralysis.[7] Furthermore, **poor T-cell-mediated immunity** is a critical risk factor, with low $CD4^+$ and $CD8^+$ T-cell levels predicting IPA risk and mortality.[8,9] In the immunocompetent host, the adaptive response to *A. fumigatus* is tightly regulated by distinct T-helper subsets: Th1 cells are essential for controlling fungal growth, whereas a skewed Th2 response can exacerbate disease. While Th17 cells facilitate neutrophil recruitment, Regulatory T cells (Tregs) play a crucial homeostatic role by downregulating inflammation to prevent organ injury and inhibit Th2-driven allergic pathology.[10] In the immunocompromised patient, this delicate balance is lost and, consequently, this multi-layered immune failure enables IPA establishment. The resulting symptoms (fever, cough and pleuritic pain)[11] are frequently masked by the underlying malignancy or treatment toxicity, causing **diagnostic delay**.[2] On CT, this angio-invasive pattern manifests as nodules with surrounding ground-glass opacity (the halo sign), which may later develop an air crescent or cavity as neutropenia resolves (see Figure 2).[2] Pathologically, invading hyphae cause haemorrhagic infarction and coagulative necrosis, the hallmark tissue damage that underlies symptoms and facilitates haematogenous spread in disseminated IA.[11]

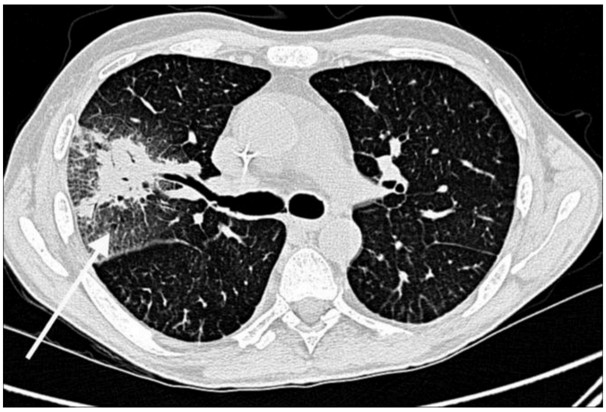

Figure 2. **CT scan of Invasive Pulmonary Aspergillosis (IPA)**. CT image from a 41-year-old man with acute myeloid leukaemia and neutropenic sepsis, showing an invasive fungal infection in the upper-right lobe. The infection appears as a mass surrounded by a peripheral halo of ground-glass opacification (arrow). *Obtained from The Lancet Infectious Diseases, Vol. 15, Schelenz et al., "British Society for Medical Mycology best practice recommendations for the diagnosis of serious fungal diseases," pp. 461–474, Copyright (2015), with permission from Elsevier.*

The incidence and mortality of IPA in this population underscore a persistent and grave clinical burden. Recent data (2021-2025) indicate that IPA affects approximately 5–6% of all AL patients, with incidence rates estimated at **5.84 per 100 patients**.[4] Despite advances in antifungal prophylaxis and diagnostics, outcomes remain dire; six-week all-cause mortality in hematology patients with IPA consistently ranges from 31% to 36%, rising to 41–50% at 12 weeks. [4]

The treatment for IPA has long relied on a limited arsenal of systemic antifungal agents, among which the **azole** class has been the mainstay of first-line therapy.[3] These drugs inhibit the synthesis of **ergosterol**, a pathway unique to fungi and yeast and a critical component of their cell membrane, and their introduction significantly improved survival outcomes.[2,3] Nevertheless, azole therapy is fraught with challenges: significant **drug-drug interactions** complicate co-administration with critical hematological agents, while **side effects** range from hepatotoxicity and visual disturbances to neurological sequelae, necessitating rigorous therapeutic drug monitoring in critically ill patients.[2]

Furthermore, **emerging resistance** to azoles threatens to undermine clinical outcomes and further narrow the already limited therapeutic arsenal.[4] Driven by the widespread agricultural use of azole fungicides and selective pressure in clinical settings, resistant *A. fumigatus* strains are proliferating globally, frequently mediated by mutations in the **cyp51A gene**, rendering first-line therapies ineffective.[3] This resistance crisis forces a reliance on alternative agents like liposomal amphotericin B or echinocandins, which are often **less effective** and carry **toxicities** that far exceed those of the azole class.[3]

This critical situation underscores the urgent need for novel strategies, where **Chimeric Antigen Receptor** (CAR) **immunotherapy**, repurposed from its remarkable success in oncology, presents a promising immunotherapeutic alternative with a precise mechanism of action.[12] The rationale for CAR therapy in IPA is twofold, directly addressing the multi-layered immune failure observed in patients with AL.[13] Firstly, the clinical imperative in these high-risk populations demands **targeted, non-cross-resistant treatments** to circumvent cumulative drug toxicity and the severe adverse effects associated with conventional systemic agents.[12] Secondly, CAR-engineered cells provide a novel, MHC-independent mechanism that directly **augments host immunity**.[13] This is particularly advantageous for AL patients, as it overcomes the limitations of traditional adoptive T-cell therapies which are often hindered by the host's underlying pancytopenia and the low frequency or inconsistent quality of endogenous *Aspergillus*-specific T cells.[13] By targeting specific antigens on the *A. fumigatus* cell wall, these engineered cells stimulate the production of **fungicidal effector molecules** and **pro-inflammatory cytokines** (such as granzymes and interferon-gamma) which in turn recruit and enhance endogenous immune clearance by activating host macrophages.[12] This represents a paradigm shift from simple pharmacologic inhibition to targeted immune activation, offering a potential solution to the dual challenges of rising fungal resistance and severe systemic drug toxicity.[14]

2. Aim of the project

2.1 Hypothesis

The hypothesis of this project is that CAR T-cells targeting *A. fumigatus* cell-wall antigens will provide a **targeted**, **MHC-independent** therapy for IPA that **bypasses the immune paralysis** seen in AL patients. By combining direct **fungicidal activity** with **endogenous immune recruitment,** this approach will significantly reduce fungal burden and improve survival outcomes compared to conventional antifungal agents.

2.2 Objectives

Therefore, the goal of this project is to provide a complete preclinical proof-of-concept for an immunotherapy-based approach against IPA using CAR technology. This goal will be pursued through four key objectives:

1. To **identify** and validate a **target** on *A. fumigatus'* surface for CAR recognition.
2. To **design** and **build** a functional CAR construct and **produce** engineered T-cells.
3. To **test** these CAR-T cells *in vitro* for their ability to specifically recognize, activate against, and eliminate the fungal target.
4. To **assess** the *in vivo* therapeutic **efficacy**, **biodistribution**, and **safety** of CAR-T cells in a relevant disease model.

3. Methodology

This section outlines the experimental workflow for the preclinical development and evaluation of CAR-T cells targeting *A. fumigatus* as a potential immunotherapeutic strategy for IPA. Figure 3 summarizes the sequential methodology, showing how each stage contributes to the achievement of the project objectives. The workflow is organized into five stages: (1) identification and validation of a suitable fungal surface target and selection of the monoclonal antibody; (2) CAR design, vector construction, and lentiviral production; (3) CAR-T cell manufacturing and expansion from human PBMCs; (4) *in vitro* functional assessment of antigen-specific activation and antifungal activity; and (5) an *in vivo* pilot study evaluating therapeutic efficacy and safety. Together, these steps provide a structured framework to generate a preclinical proof of concept for CAR-T cells capable of specifically recognizing, activating against, and eliminating *A. fumigatus*.

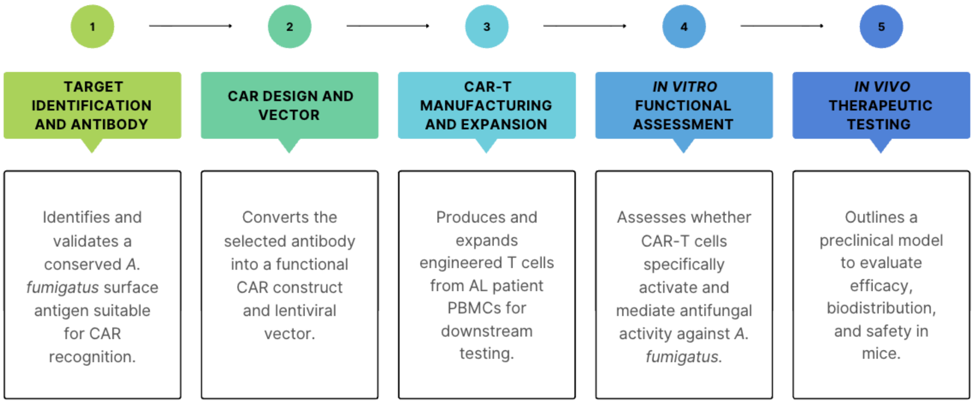

Figure 3. Workflow for the preclinical development and evaluation of antifungal CAR-T cells targeting *Aspergillus fumigatus*. *Created using Canva*

Each stage corresponds to a work package (WP) designed to ensure a seamless transition from molecular engineering to functional validation. To guarantee the feasible execution of these interdependent phases, milestones and deliverables have been strategically aligned across the project's duration. This integrated timeline is summarised in Figure 4, which presents a trimester-based chronogram of the full three-year experimental plan.

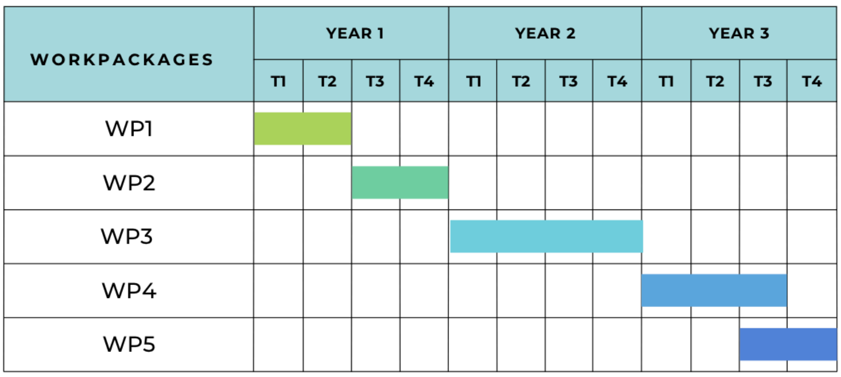

Figure 4. Three-year trimester-based chronogram of the experimental methodology. Each row represents one of the five methodology sections, and colored bars indicate the trimesters dedicated to each phase, from target identification through *in vivo* testing. *Created using Canva*

3.1 Target identification and antibody

The first step in designing an effective CAR-T cell therapy against IPA will be the identification of a suitable target antigen on the surface of *A. fumigatus*. The ideal antigen must be a **conserved**, **accessible** molecule on the fungal cell wall that is **constitutively expressed**[15,16] on both conidia and, critically, on the hyphal form responsible for invasive tissue damage.[17] Moreover, to guarantee **wide recognition** and **minimise off-target effects**, the protein should be stably produced across clinical isolates with limited resemblance to human proteins.[15,18]

For this project's CAR design, the **monoclonal IgG1 antibody AB90-E8**[14] will be selected as the targeting domain. This antibody recognises a conserved surface protein epitope on *A. fumigatus* hyphae, strongly decorates early germinating hyphae from multiple clinical isolates, and does not bind other pathogenic *Aspergillus* species.[14] Furthermore, AB90-E8 has been previously characterised for its ability to bind the fungal cell wall and mediate antifungal activity, and has already served as the basis for a CAR construct in preclinical models of IPA, where AB90-E8-based CAR-T cells reduced fungal burden and improved survival.[14] This prior validation makes AB90-E8 a rational and translationally relevant choice.

While alternative antibodies such as **1D2**, which targets a hyphal cell-wall glycoprotein and broadly inhibits several clinically important *Aspergillus* species,[19] could enable a more pan-*Aspergillus* CAR in future work, the focus of this project on *A. fumigatus*-specific IPA and the existing preclinical validation of an AB90-E8-based CAR as the primary targeting domain.

3.2 CAR design and vector

To maximize therapeutic efficacy against *A. fumigatus* in IPA, the CAR will be engineered based on a fourth-generation design, known as **TRUCK** (T cells redirected for universal cytokine-mediated killing).[20] In contrast to previous CARs, fourth-generation designs incorporate an **antigen-inducible cytokine cassette**.[20] This modification enables the recruitment and activation of innate immune effectors cells, that are critically deficient in patients with neutropenic IPA.[21] By bolstering the local innate immune response, this platform offers a MHC-independent therapeutic strategy.[22,23] Furthermore, its potential to enhance immune cell recruitment and persistence aligns with the project's central objective and is supported by emerging evidence from recent studies in fungal models.[14,23]

From this point onward, the fourth-generation antifungal CAR described in this thesis will be referred to as **Fumi-CAR**, a graphical representation of which is provided in Figure 5.

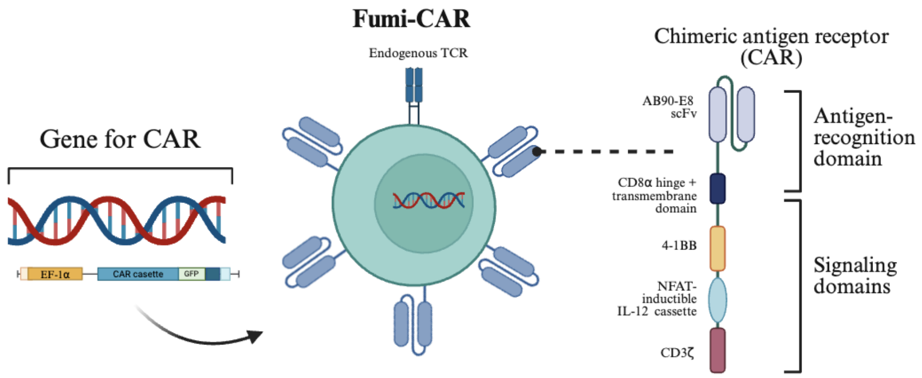

Figure 5. Fourth-generation TRUCK design for antifungal CAR T-cells. The lentiviral construct (left) contains an EF-1α promoter driving a CAR cassette linked via T2A to a GFP reporter. The chimeric antigen receptor (right) comprises an AB90-E8-derived scFv for antigen recognition, a CD8α hinge and transmembrane domain, a 4-1BB costimulatory domain, a CD3ζ signaling domain, and an NFAT-inducible IL-12 cassette, enabling antigen-dependent cytokine production. *Created using Biorender.*

The **extracellular domain** will be a single-chain variable fragment (scFv) from the AB90-E8 monoclonal antibody, which binds a conserved protein epitope on *A. fumigatus* hyphae with high affinity (K_D ~10 nM), as explained in section 3.1. The scFv format (V_H–$(Gly_4Ser)_3$–V_L) ensures monomeric expression and avoids Fc-mediated effects, with sequences codon-optimized for human expression.[14]

The **spacer** will use the human **CD8α hinge** (45 aa) and transmembrane domain (21 aa), selected for its short, rigid structure that facilitates access to dense hyphal antigens while promoting stable dimerization and surface expression (~80–90% in T cells).[14]

The CAR construct will utilize the **4-1BB** (CD137) intracellular domain for **co-stimulation**, a choice predicated on its superior integration with transgenic cytokine payloads and its capacity to sustain T-cell fitness in the immunosuppressive microenvironment of AL patients.[24,25] Recent evidence indicates that while CD28 drives rapid effector differentiation, 4-1BB signaling facilitates metabolic plasticity by favoring **mitochondrial biogenesis and oxidative phosphorylation** (OXPHOS).[24] This metabolic profile effectively **mitigates the upregulation of exhaustion markers** like PD-1 and LAG-3,[24,25] ensuring that the CAR-T cells do not reach terminal differentiation before clearing complex *Aspergillus* hyphal structures. In the context of invasive pulmonary aspergillosis (IPA), 4-1BB provides a tempered activation threshold that prevents the hyper-inflammatory cytokine bursts often seen with CD28 CARs.[26,27] Furthermore, 4-1BB-based constructs have demonstrated a unique capacity to maintain effector function despite the lymphopenia and corticosteroid exposure common in AL treatment protocols.[25,28]

The key innovation will consist of an **NFAT-inducible IL-12 cassette**.[20] IL-12 is the optimal cytokine for this antifungal TRUCK, as it is the most robustly validated mediator of IPA protection.[23,29] It is established that IL-12 is **essential for Th1 differentiation**, **IFN-γ production**, and **macrophage fungicidal activity** against *A. fumigatus*,[30] functions critically impaired in neutropenic AL patients.[31] Unlike other candidates, IL-12 directly **enhances human phagocyte oxidative burst** and **hyphal killing**, and IL-12 knockout mice exhibit dramatically increased IPA susceptibility.[23] Furthermore, recombinant IL-12 or IL-12-engineered DCs reduce fungal burden and improve survival in murine

IPA, confirming its capacity for Th1 reconstitution in immunocompromised hosts.[32] The IL-12 cassette is placed under an NFAT-inducible promoter so that IL-12 is produced only upon T-cell activation, a strategy shown to enhance therapeutic efficacy while keeping systemic IL-12 levels undetectable and thereby minimizing cytokine-related toxicity.[33]

Finally, primary signaling will be delivered by human **CD3ζ** (112 aa, 3 ITAMs), initiating proximal signaling (ZAP-70, LAT) upon scFv crosslinking.[14]

The full cassette (scFv-CD8α-4-1BB-CD3ζ-IL12) is cloned into a **third-generation lentiviral vector** for enhanced biosafety[34] via split packaging (psPAX2/pMD2.G), eliminating replication-competent lentivirus risk.[35] The **EF-1α promoter** ensures stable T-cell expression[36] while a **GFP reporter** is included via T2A for transduction tracking.[37]

Lentivirus production uses standard **HEK293T** co-transfection: change of media at 18h post transfection and harvest at 72/96h, filtration, and centrifugation, targeting 10^8–10^9 TU/mL titers quantified on HEK293T.[35,38]

3.3 CAR-T manufacturing and expansion

To mimic clinical settings and assess viability in a cytokine-altered, lymphopenic setting, Fumi-CAR cells will be generated from the PBMCs of AL patients. With informed consent and ethics committee approval for genetic modification and research use, all operations will adhere to an approved research protocol.

3.3.1 Collection and isolation of PBMCs

PBMCs will be initially isolated by **leukapheresis**, the standard approach for clinical CAR-T manufacturing in hematologic malignancies. The leukapheresis product (about 150–200 ml) will be collected in anticoagulant (ACD-A) and processed within 24 h.[39]

PBMCs will be initially isolated by density-gradient centrifugation using Ficoll-Paque™.[40] Subsequently, $CD3^+CD56^-$ T cells will be purified from the PBMC fraction using **fluorescence-activated cell sorting** (FACS). To minimize contamination with leukaemic blasts, additional exclusion markers will be included as necessary.[41] The purified T cells will be resuspended in TexMACS™ Medium and counted using trypan blue exclusion; a viability of ≥90% will be required to proceed with manufacturing.[40,41]

Employing FACS at this stage, rather than relying on positive selection with magnetic beads, improves purity and is particularly advantageous when the starting frequency of healthy T cells is low or the lymphocyte population is distorted, as is often the case in AL.[42]

3.3.2 T-cell activation and transduction

Following purification, the T cells will be activated and genetically modified to express the Fumi-CAR. Sorted $CD3^+$ cells will be stimulated with anti-CD3/4-1BB antibody-coated beads in a specialized TexMACS™ Medium supplemented with human serum.[41,43]

Throughout the activation and culture period, the medium will be supplemented with MACS® Cytokines Human **IL-7** (40 IU/mL) and Human **IL-15** (40 IU/mL). The combination of IL-7 and IL-15 is the optimal cytokine regimen because it reconciles the temporal paradox of the disease: it generates a T-cell product with **immediate effector function** sufficient to control acute fungal invasion, while simultaneously programming the metabolic fitness and epigenetic landscape required for **long-term persistence**, thereby providing coverage throughout the unpredictable duration of post-chemotherapy immunodeficiency.[13,41]

Activated T cells will be transduced with the lentiviral vector encoding CAR construct using a **spinoculation** method. Spinoculation, which employs centrifugation to enhance virus-particle contact with target cells, is the ideal technique for maximizing transduction efficiency, especially when starting cell numbers are limited.[44]

3.3.3 Expansion and characterisation

After transduction, the CAR-T cells will be expanded and subjected to rigorous quality control to confirm their phenotype and functionality before proceeding to *in vitro* assays. Cultures will be maintained in TexMACS™ Medium with MACS® Cytokines Human IL-7 (40 IU/mL) and Human IL-15 (40 IU/mL), with cell density monitored and adjusted according to established protocols to support optimal growth.[40,41]

A comprehensive phenotypic and functional characterisation will then be performed:

- **Transduction Efficiency**: This will be quantified by flow cytometry, detecting the GFP reporter gene that was incorporated into the lentiviral vector design. [41]
- **T-Cell Phenotype**: The CD4/CD8 ratio and the composition of memory subsets will be analyzed by flow cytometry using markers such as CD45RA, CCR7, and CD62L. This analysis will confirm whether the IL-7/IL-15 support has successfully enriched cells with a central memory phenotype, as intended.[41]
- **Exhaustion Marker**s: The surface expression of exhaustion markers, including PD-1, CD39 and TIM-3, will be evaluated. This assessment will help verify whether the 4-1BB costimulatory domain incorporated into the CAR design contributes to a less exhausted phenotype.[40]

This thorough characterisation ensures that the Fumi-CARs are appropriately transduced, viable, and phenotypically suited for the planned functional assays.

3.4 *In vitro* functional assessment

In vitro assays will determine whether Fumi-CAR cells derived from AL patients are capable of recognizing *A. fumigatus*, becoming activated, exerting direct antifungal effects, and secondarily activating macrophages. These experiments are essential to test the central hypothesis that CAR engineering can restore and amplify antifungal immunity in an otherwise functionally paralysed immune system.

Fumi-CAR cells and matched non-transduced patient T cells as a negative control will be co-cultured with *A. fumigatus* hyphae to assess antigen-dependent activation. The co-culture (see Figure 7A) will be designed so that:

- Hyphae are generated from conidia in a defined medium over 16–24 hours, ensuring a mature filamentous phase that mimics the invasive form present in IPA.[13]
- Fumi-CAR and non-transduced T cells from the same AL donor are added at controlled effector:target ratios and incubated for 16–24 hours to allow for receptor engagement and early effector responses.[13]
- Conditions without hyphae are included to quantify background activation and cytokine secretion in the absence of fungal antigen.

3.4.1 Antigen-specific activation and cytokine release

From these co-cultures, two main readouts will be obtained (see Figure 7B):

- **Cytokine profile**: Supernatants will be analysed for IFN-γ, TNF, GM-CSF, IL-2, IL-10 and IL-12 using ELISA.[45] Increased production of Th1-type cytokines and detectable IL-12 specifically in Fumi-CAR + hyphae conditions, compared with non-transduced T cells, will indicate that the TRUCK design is functioning as intended and that fungal recognition triggers a pro-inflammatory programme consistent with antifungal protection.
- **T-cell activation markers**: Cells will be stained for and analysed by flow cytometry. A higher proportion of $CD69^{+}CD25^{+}$ cells among Fumi-CAR than among non-transduced T cells upon exposure to hyphae will confirm antigen-specific activation.[46] These data will also allow Fumi-CAR contributions to the response.

Should the TRUCK design function as intended, Fumi-CAR cells will show significantly higher $CD69^{+}CD25^{+}$ expression and elevated IFN-γ, TNF, GM-CSF, and IL-12 production compared to non-transduced T cells, confirming restoration of antigen-specific activation in the immunocompromised AL setting.

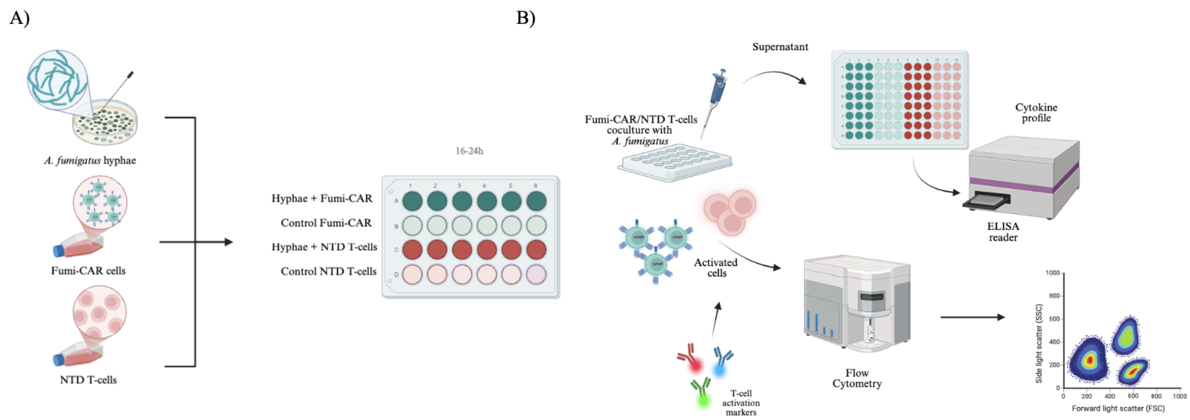

Figure 7. Assessment of Fumi-CAR T-cell functional activation. (A) Experimental co-culture design. *A. fumigatus* hyphae were co-cultured with either Fumi-CAR or non-transduced (NTD) T-cells from the same donor. Control conditions included T-cells cultured without hyphae. **(B) Antigen-specific activation and cytokine release**. Following co-culture, supernatants were analyzed by ELISA (IFN-γ, TNF, GM-CSF, IL-2, IL-10, IL-12), and cells were stained for flow cytometry to evaluate activation markers ($CD69^{+}$, $CD25^{+}$). *Created using BioRender*

3.4.2 Hyphal damage and fungal burden

To link activation to functional antifungal activity, two complementary assays will be used: XTT for hyphal damage and qPCR for fungal burden.

- **Hyphal damage (XTT assay):** After co-culture with Fumi-CAR or non-transduced T cells, hyphae will be incubated with XTT reagent. Reduced metabolic activity (lower colour development) in Fumi-CAR conditions compared with controls will indicate increased hyphal damage. This provides a relatively rapid, quantitative measure of fungal viability.[47]
- **Fungal burden (qPCR):** In matched wells, total DNA will be extracted after co-culture and fungal DNA quantified by qPCR using an *A. fumigatus*-specific target. A lower fungal DNA signal in Fumi-CAR co-cultures than in non-transduced T-cell conditions will demonstrate that CAR engineering not only damages hyphae metabolically but also reduces overall fungal biomass. Using qPCR avoids variability inherent to plating.[46]

A reduction in XTT metabolic activity and lower fungal DNA signal in Fumi-CAR co-cultures compared to non-transduced T cell controls would confirm that antigen recognition translates into measurable direct antifungal killing.

3.4.3 Macrophage activation assay

To assess the TRUCK concept of recruiting residual phagocytes, human monocyte-derived macrophages will be added to co-cultures of hyphae with Fumi-CAR or non-transduced T cells.[48] Readouts will include:

- Upregulation of macrophage activation markers such as HLA-DR, CD80 and CD86 by flow cytometry.[48,49]
- Changes in cytokines associated with macrophage activation, for instance TNF, IL-1β and IL-12 in the supernatant.[48,50]

Upregulation of HLA-DR, CD80, and CD86 on macrophages alongside increased TNF and IL-12 in the supernatant, specifically in Fumi-CAR conditions, would validate the IL-12-mediated TRUCK mechanism and support the hypothesis that engineered cytokine release can recruit residual innate immunity in an AL-associated immunocompromised context.

3.5 *In vivo* therapeutic testing

To validate the antifungal efficacy of Fumi-CAR cells in a physiologically relevant setting, an initial *in vivo* study will be conducted in immunocompromised **NSG mice** with IPA, based on established murine models of the disease.[13,14]

3.5.1 Mouse model and treatment groups

As illustrated in Figure 8, NSG mice (defective T- and B-cell immunity) will be additionally immunosuppressed with **cyclophosphamide** to mimic chemotherapy-induced neutropenia and infected intratracheally with 1.5×10^8 *A. fumigatus* conidia to establish IPA in a manner that resembles the clinical situation of AL patients, as opposed to intravenously.[13,14]

Each experimental group will comprise eight animals; four male and four female, to account for potential sex-dependent differences in immune responses and to confirm that therapeutic efficacy is independent of sex, in line with current preclinical research standards.[52] Three groups will be included: vehicle control, non-transduced human T cells, and Fumi-CAR cells, for a total of 24 animals.

Fumi-CAR cells or matched control T cells will be administered intravenously at 5×10^6 cells per injection, delivered at 6 and 48 hours post-infection, following the dosing schedule validated in a comparable NSG cyclophosphamide model of IPA.[13] This two-dose regimen ensures sufficient CAR-T engraftment during the critical early phase of fungal invasion, in a therapeutic rather than prophylactic scenario.

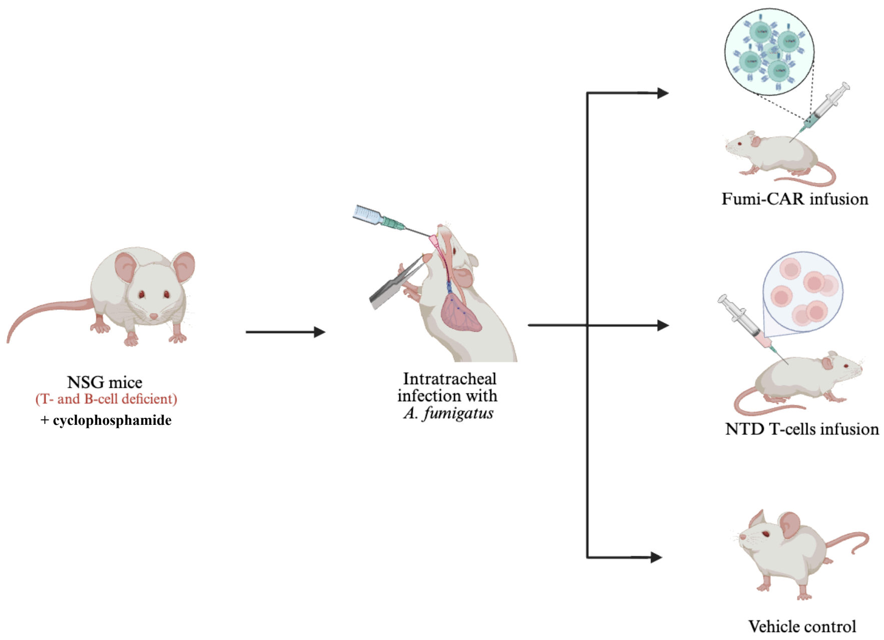

Figure 8. *In vivo* therapeutic model in NSG mice. NSG mice are immunosuppressed with cyclophosphamide to mimic chemotherapy-induced neutropenia, then infected intratracheally with *A. fumigatus* conidia. Mice are randomized to receive vehicle control, non-transduced human T-cells, or Fumi-CAR cells to evaluate therapeutic efficacy. *Created using Biorender*

3.5.2 Outcome measures

Key outcome measures will include:

- **Survival**: Kaplan–Meier curves over 14 days.[14]
- **Fungal burden**: qPCR in lung homogenates.[14]
- **Histopathology**: lung sections stained to assess inflammation, hyphal invasion and tissue damage.[14]
- **CAR-T distribution**: Human $CD45^+$ cells (surrogate for Fumi-CAR T cells) would be quantified in extracted lung and blood samples post-infection, using a human-specific CD45 antibody that does not cross-react with murine $CD45^+$ cells. Non-relevant tissues not involved in IPA would serve as controls to confirm specific localisation to the infection site.[51]
- **T cell activation** (*ex vivo)*: Ca^{2+} flux in Fumi-CAR cells measured by dynamic imaging microscopy in healthy vs. *A. fumigatus*–infected human lung tissue slices.[14]

Taken together, these endpoints are expected to demonstrate improved 14-day survival, reduced pulmonary fungal burden, preserved alveolar architecture, and specific CAR-T localisation to the lung in Fumi-CAR-treated animals, with consistent outcomes across both sexes.

3.6 Intellectual Protection, Scientific Dissemination, and Public Communication

Should the Fumi-CAR platform yield promising preclinical results, a coordinated strategy encompassing intellectual protection, scientific publication, and public communication will be pursued.

Prior to any dissemination, a **patentability analysis** will be conducted in collaboration with the hospital's technology transfer office to assess whether the CAR construct, manufacturing protocol, or any novel findings warrant intellectual property protection. If so, a patent application will be filed before results are made public, ensuring that the institution retains rights over potential clinical or commercial translation.

Scientific results will subsequently be submitted for publication in peer-reviewed, **open-access journals**, prioritising outlets of high visibility within the fields of immunology, mycology, and translational medicine; such as *mBio*, *Journal of Fungi*, or *Frontiers in Immunology*, so that findings are freely accessible to the global research community without paywall restrictions. Preprint deposition on platforms such as *bioRxiv* will also be considered to accelerate knowledge sharing ahead of formal peer review. Additionally, results will be presented at national and international conferences in infectious disease and cell therapy.

In parallel, a **non-specialised communication strategy** will be implemented to ensure that the discovery reaches patients, families, and the broader public. This will involve collaboration with patient associations, particularly the *Fundació Josep Carreras contra la Leucèmia*, given its focus on leukaemia patients who constitute the primary target population of this therapy. Accessible plain-language summaries of key findings will be prepared and disseminated through hospital outreach channels, including institutional websites, newsletters, and public awareness events. To centralise this effort, a dedicated public outreach website has been designed to communicate the project's aims, approach, and expected impact in accessible, jargon-free language for patients, families, and the general public. The goal is to ensure that those most affected by IPA and leukaemia are informed of emerging therapeutic possibilities in a transparent and comprehensible manner.

4. Limitations, Expected Impact, and Ethical and Biosecurity Considerations

4.1 Limitations of the project

This project presents several inherent limitations that must be acknowledged to contextualise its scope and feasibility. A primary constraint arises from the use of **PBMCs derived from AL patients**, which are frequently limited in quantity and may be functionally compromised. This can affect both the efficiency of cell sorting and the reproducibility of the downstream manufacturing steps. Moreover, the intrinsic poor fitness, exhaustion, or skewed subset distribution of T cells from heavily pre-treated AL patients may fundamentally limit expansion and potency, even with IL-7/IL-15 support.[13,31] In addition, patient-to-patient variability is expected to be considerable, since prior chemotherapy, corticosteroid exposure and the degree of lymphopenia will influence the phenotype and proliferative capacity of the cells.[31, 53]

The **engineering strategy** also introduces technical constraints. Fumi-CAR combines antigen recognition, co-stimulation, inducible cytokine release and a fluorescent reporter in a single construct, which increases the complexity of vector design and validation.[20,34] Any stage of this process may become a bottleneck: lentiviral production may yield variable titres, transduction may be inefficient despite spinoculation, and the final product may show heterogeneous expression of CAR and GFP.[34,35] Furthermore, **reliance on a single antibody (AB90-E8)** that recognizes one conserved protein epitope on *A. fumigatus* hyphae introduces the risk of antigen escape or target downregulation. Under immune pressure from Fumi-CAR cells, fungal variants with reduced expression of the target epitope, or transcriptional adaptation during different growth phases (e.g., conidia vs. hyphae), could emerge and escape recognition.[14] This phenomenon is well-documented in cancer CAR-T therapy and represents a plausible resistance mechanism that this proof-of-concept project does not address.[15,16] Future iterations may require multi-target CARs (e.g., recognizing two distinct fungal antigens) to mitigate this risk.

The proposed ***in vitro* assays**, while robust for proof-of-concept validation, remain reductionist models that cannot fully replicate the pulmonary microenvironment of invasive pulmonary aspergillosis. Fungal co-cultures and macrophage activation experiments capture key aspects of antifungal activity but omit vascular invasion, tissue architecture, and multi-cellular interactions present *in vivo*.[13,14] What's more, the **NFAT-inducible IL-12 cassette** is designed to restrict cytokine production to the site of CAR activation, minimizing systemic toxicity. However, the potential for local pulmonary inflammation or cytokine-mediated immunopathology at high fungal burdens remains unknown prior to *in vivo* testing, and would require careful dose-escalation and monitoring in the proposed murine model.[33] Consequently, the pilot *in vivo* study represents an initial validation, with extended studies required for full preclinical development.

Finally, translational barriers include the **high cost and technical demands** of personalised CAR-T manufacturing, alongside the need for stringent biosafety oversight (see 4.3 section).[34,43] These factors position the project as a foundational platform requiring substantial refinement before clinical applicability.

Table 1 provides a structured summary of the principal methodological risks identified across all sections of this thesis; from target identification through to translational considerations, together with their potential consequences and corresponding mitigation strategies. Several of these mitigations have already been incorporated into the methodology described above; the table serves to consolidate them and enhance the overall feasibility and reproducibility of the project.

Table 1. Methodological limitations, risks, and mitigation strategies

Manufacturing & Cell Source

Limitation	Possible consequence	Mitigation
Low cell yield and viability from AL patient leukapheresis	Insufficient starting material; reduced reproducibility	Define minimum release thresholds; process promptly; run healthy-donor pilots first [39,40]
T-cell functional impairment, exhaustion, and corticosteroid-driven suppression	Poor expansion, low transduction efficiency, manufacturing failure	IL-7/IL-15 supplementation; strict quality-control release criteria throughout
Donor-to-donor variability in chemotherapy exposure, lymphopenia, and blast contamination	Inconsistent outcomes; reduced statistical interpretability	Paired donor comparisons; per-donor analysis; increased biological replicates [13,31,53]

Target & Antigen

Limitation	Possible consequence	Mitigation
Reliance on a single, molecularly uncharacterised epitope on hyphae	Antigen escape under immune pressure; reduced cross-isolate recognition	Surfaceome profiling to identify target; validate across clinical isolates; consider multi-target CARs in future iterations [14,15]
AB90-E8 does not recognise resting or swollen conidia	Therapy ineffective during early pre-germination infection stages	Administer once hyphal growth is established; combine with early antifungal prophylaxis to cover the conidial phase [14]

CAR Design & Engineering

Limitation	Possible consequence	Mitigation
High construct complexity combining scFv, 4-1BB, IL-12 cassette, and GFP in a single vector	Cloning failure, variable titres, heterogeneous expression	Stepwise validation; sequence-confirm each cloning stage; test modules individually before full assembly [34]
NFAT-inducible IL-12 may cause local pulmonary immunopathology at high fungal burdens	Tissue damage confounding therapeutic benefit interpretation	Dose-ranging studies; monitor cytokine release and inflammatory markers at all *in vivo* endpoints [29,33]

***In vitro* Assessment**

Limitation	Possible consequence	Mitigation
Co-culture systems omit vascular invasion, tissue architecture, and multi-cellular *in vivo* interactions	Overestimation of antifungal efficacy that may not translate *in vivo*	Complement with *ex vivo* human lung tissue slice Ca^{2+} flux assay; frame results as proof-of-concept only [13,14]

***In vivo* Model**

Limitation	Possible consequence	Mitigation
Possible failure to establish consistent IPA following intratracheal inoculation	Absence of infection in some animals; confounded survival and fungal burden data; inability to assess therapeutic efficacy	Monitor clinical signs using the modified murine sepsis score (MSS) at 6 hours post-infection before proceeding to CAR-T infusion; only include mice meeting predefined infection severity thresholds; confirm fungal invasion histopathologically at endpoint in all groups [13, 54]
Fourteen-day follow-up does not capture long-term persistence, late toxicity, or infection relapse	Incomplete safety and efficacy profile	Frame explicitly as pilot study; plan extended follow-up cohorts; quantify residual CAR-T cells at endpoint [13,14]

4.2 Expected impact of the project

Success in this project would yield significant value across clinical, scientific, and societal domains. Clinically, Fumi-CAR addresses a critical unmet need in AL patients with IPA, where neutropenia, T-cell dysfunction, and antifungal resistance limit current options. By enabling direct hyphal recognition and localised IL-12-mediated innate immune recruitment, the platform could reduce fungal burden, attenuate tissue invasion, and improve survival in this high-mortality population.

Scientifically, the work would extend CAR technology beyond oncology, validating TRUCK designs for infectious disease and generating insights into fungal antigen recognition paired with 4-1BB/IL-12 signalling in lymphopenic settings. This knowledge would inform broader applications in antifungal immunotherapy and immune engineering.

Societally, improved outcomes for leukaemia patients with IPA would alleviate the emotional and logistical burden on families and healthcare systems, while reducing reliance on prolonged antifungal regimens. Educationally, the project exemplifies interdisciplinary integration of microbiology, immunology, and cell therapy, demonstrating how targeted immune restoration can address pathogen- and host-specific challenges.

Economically, initial implementation would be costly due to individualised manufacturing. However, a modular platform adaptable to other fungal targets or immune deficits holds long-term potential for cost efficiencies through scale-up and reuse. Ultimately, Fumi-CAR represents a step toward personalised medicine, where therapies are reprogrammed for specific biological contexts, paving the way for flexible immune interventions across diseases.

4.3 Ethical and biosecurity considerations

The project engages human biological material, genetic modification, and a pathogenic fungus, necessitating rigorous ethical and biosafety frameworks. Patient-derived PBMCs require informed consent detailing study aims, sample use, and data anonymisation, with approval from the local ethics committee. As a vulnerable population, AL patients warrant particular scrutiny to ensure voluntary participation and minimal donor burden. A copy of the informed consent form is provided in the Appendix of this thesis.

Biosafety protocols must govern lentiviral vector production and handling under BSL-2 conditions, with strict aerosol containment, waste decontamination, and segregation of pre- and post-transduction materials.[34] *A. fumigatus* manipulation similarly demands controlled laboratory practices to protect personnel and prevent environmental release.[55]

Broader dual-use concerns arise from engineered immune cells and viral vectors, which must remain confined to therapeutic intent under institutional oversight. Governance includes biosafety committee review, training in good laboratory practices, and transparent reporting of results. Should the platform advance, regulatory preclinical studies, GMP compliance, and clinical trial ethics would be mandatory prerequisites.

These considerations are not obstacles but integral to responsible innovation, ensuring that potential benefits outweigh risks while upholding scientific integrity and patient safety.

5. Budget plan

The budget for the project is estimated considering that it would be carried out in a hospital, where the personnel is already hired and basic laboratory equipment such as centrifuges, biosafety cabinets, incubators, and flow cytometers are available. The team would consist of a principal investigator, a laboratory technician dedicating part of their time to the project, a researcher with expertise in immunology and fungal infections, and a doctoral or predoctoral researcher. Personnel would be selected according to scientific merit and suitability for the project, regardless of sex, gender, origin, or sexual orientation.

The budget is broken down below for a project duration of approximately 3 years.

- **Personnel**: 0€ (salaries covered by the institution or external grants).
- **Establishment and infrastructure**: 0€ (basic equipment available at the centre).
- **External services** (cell sorting, histopathology, animal facility): 24,000€. This includes approximately 13,000€ for animal facility services, covering the purchase of 24 NSG mice at subsidised institutional rates (~200€/mouse), specialised housing (~25€/cage/week for 8 cages over 8 weeks), and all associated procedures (cyclophosphamide administration, intratracheal infection, intravenous injections, and endpoint analyses). The remaining ~11,000€ covers cell sorting, histopathological processing, and imaging services.
- **Consumables, antibodies, reagents, lentiviral production materials, and cell culture supplies**: 55,000€.
- **Equipment maintenance** (flow cytometer, qPCR thermocycler, ELISA plate reader, biosafety cabinet, centrifuges, CO_2 incubators, −80°C freezer, and fluorescence microscope): 10,000€, *institutional service contracts partially subsidised; figure reflects project-attributable share.*
- **Conference attendance** (2,000€/year): 6,000€.
- **Open-access publication costs** (~3,000€/article, estimated 2 publications): 6,000€.
- **Patentability analysis and patenting**: 13,000€.

Direct costs subtotal: 114,000€. An additional 25% is added for indirect costs and unforeseen events (**28,500€**), resulting in a **total estimated budget of 142,500€.**

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7. APPENDIX

Informed Consent Form

We are asking you to participate in a research study. This form is designed to give you information about this study. We will describe this study to you and answer any of your questions.

Project Title: Fumi-CAR. A Chimeric Antigen Receptor Immunotherapy Approach to Eradicate *Aspergillus fumigatus* in the Immunocompromised Leukaemic Lung

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KEY INFORMATION

Provided below is a short summary of this study to help you decide whether you want to participate. More detailed information is listed later in this form.

- The purpose of this research is to develop a preclinical CAR-T cell treatment targeting *Aspergillus fumigatus* for patients with acute leukemia. To do so, we will obtain blood-derived immune cells from acute leukemia patients and study how they can be used to fight fungal infections.
- You will be asked to donate a blood sample that will be used to isolate immune cells for laboratory research procedures.
- Your participation in the study is expected to take approximately 5–10 minutes, which is the time needed for the blood draw.
- No specific risks are expected beyond those rarely associated with a routine blood draw. These may include mild and short-lasting discomfort at the puncture site, slight bruising or bleeding, or, rarely, light-headedness.
- You will not receive a direct medical benefit from participating, but your participation may contribute to future scientific knowledge and to the development of new therapies for patients with acute leukemia and invasive fungal infections.
- Your participation in this research is voluntary. You may refuse to participate or withdraw at any time, without penalty and without any effect on your medical care or the relationship with your healthcare team.

DETAILED INFORMATION

What the study is about

The purpose of this research is to develop a preclinical CAR-T cell treatment targeting *Aspergillus fumigatus* for patients with acute leukemia. To do so, we will obtain blood-derived immune cells from acute leukemia patients and study how they can be modified and used to fight fungal infections. Your sample will only be used for this research project.

What we will ask you to do

We will ask you to donate a blood sample. The sample will be collected by trained healthcare personnel using standard clinical procedures. The amount of blood collected will be limited to what is needed for the research, and no additional invasive procedures are expected. Your sample will be used to isolate peripheral blood mononuclear cells (PBMCs), which may then be used for cell culture, experimental manipulation, and functional assays related to CAR-T research. The total time of your participation will be approximately 5–10 minutes, which is the time needed for the blood draw.

Risks and discomforts
No specific risks are expected beyond those rarely associated with a routine blood draw. These may include mild and short-lasting discomfort at the puncture site, slight bruising or bleeding, or, rarely, light-headedness.

Benefits
You will not receive any direct benefit from taking part in this study. However, the information obtained may help scientists better understand how to develop new antifungal CAR-T strategies and may benefit other people now or in the future.

Incentives for participation
You will not receive payment, compensation, or extra credit for being in this study.

Use of Tissue Samples/DNA for Future Studies
Your blood sample will be used for the preclinical CAR-T research project described in this consent form.

If there is any remaining material after completing this project, we would like to store it in a coded research collection to carry out future studies related to:

- Antifungal CAR-T strategies
- Immune responses in acute leukemia and other hematologic diseases
- Fungal infections, especially those caused by *Aspergillus fumigatus*

The stored material will be identified with a code, and your personal identifying information will not be kept together with the sample. Access to the code that links the sample to your identity will be restricted to the Principal Investigator, according to applicable Spanish and European regulations on biomedical research and data protection.

The samples in this collection may be used in future research projects on the topics described above, as long as these projects have been approved by the corresponding Research Ethics Committee and comply with current legislation. Your sample will not be used for research on unrelated diseases or purposes without asking for your consent again.

Stored samples may be kept for up to 5 years. You may withdraw your permission for storage and future use at any time. If you withdraw, any identifiable stored material that has not already been used in completed analyses will be destroyed according to institutional procedures and current regulations.

This research will not involve whole genome sequencing, and no immortalized cell line will be created from your sample. Specimens collected from you for this study and/or information derived from them will not be used to generate commercial profit, and you will not receive any financial benefit or other compensation from products developed using these specimens.

Your choice about storage and future use
Do you give us permission to store any remaining blood sample in a coded research collection and to use it in future projects related to antifungal CAR-T strategies, immune responses in acute leukemia and other hematologic diseases, and fungal infections such as those caused by *Aspergillus fumigatus*?

Keep in mind that you do not have to give permission for this storage and future use to participate in the main study.

☐ Yes, I agree to allow storage and future use of my remaining sample for the type of research described above.
☐ No, I do not agree to storage and future use of my remaining sample.

Signature:

Date: __/__ /____

If you are injured by this research
If any research-related activity results in injury, appropriate first aid or follow-up care will be made available. The costs of such care will be billed in the ordinary manner to you or your insurance company. No reimbursement, compensation, or free medical care is offered by the research team. If you think you have suffered a research-related injury, contact the principal investigator right away at 1626084@uab.cat.

Privacy/Confidentiality
Your personal data and your participation in this study will be treated confidentially in accordance with current European and Spanish data protection laws. Your blood sample and associated research data will be identified with a code, and your name or other direct identifiers will be stored separately from the research data. Only authorized members of the research team and, where appropriate, the institution's data protection and quality-control personnel will be able to link the code to your identity.

Electronic data will be stored on secure, access-restricted devices or servers, and physical documents will be kept in locked storage areas. When results are published or shared with other researchers, they will be presented in a way that does not allow you to be directly identified.

If your sample is stored in the research collection described above, it will be kept in coded form in accordance with Spanish regulations on biological sample collections for research. Any future projects using samples from the collection will require prior approval by the relevant Research Ethics Committee and will use coded data, without direct identification of donors in publications or communications of results.

The collected biological samples will be stored for a maximum of 5 years, and any personal data obtained will be kept for the same period (up to 5 years)

Information about use of your biospecimens

Specimens collected from you for this study and/or information derived from your specimens will not be used to generate commercial profit. You will not share in any commercial value or other compensation from products developed using these specimens.

You will not receive individual clinical results from the analyses performed on your sample, because this is a preclinical research study and the tests carried out are not designed or validated for clinical diagnosis or treatment decisions.

However, you may request general information about the overall results of the study once it has been completed, and, whenever possible, the research team will make a summary of the main findings available to interested participants.

Taking part is voluntary
Your participation is voluntary. You may refuse to participate before the study begins, stop participating at any time, or skip any procedure that you do not want to complete. Choosing not to participate will not affect your medical care, your benefits, or your relationship with the hospital or research team.

Withdrawal by investigator, physician, or sponsor
The investigators or physicians may stop your participation in the study at any time if they believe it is in your best interest, if a medical problem arises, or if the study procedures are no longer appropriate for you.

If you have questions
The main researcher conducting this study is Sònia Serarols. Please ask any questions you have now. If you have questions later, you may contact her at 1626084@uab.cat.

If you have any questions or concerns regarding your rights as a subject in this study, you may contact the relevant Institutional Review Board / Research Ethics Committee.

You will be given a copy of this form to keep for your records.

Statement of Consent
I have read the above information, and have received answers to any questions I asked. I consent to take part in the study.

Your Signature____ Date____________

Your Name (printed)____

Signature of person obtaining consent______________________________ Date____________

Printed name of person obtaining consent_________ Sònia Serarols Ylla-Català _______________

This consent form will be kept by the researcher for at least three 3 years beyond the end of the study.